

Point-by-Point Response to Reviewers

“Sparse autoencoders reveal organized biological knowledge
but minimal regulatory logic in single-cell foundation models:
a comparative atlas of Geneformer and scGPT”

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We are grateful to all three reviewers for their detailed, constructive comments. In this revision we have fixed several factual errors regarding model architecture and training objectives, softened over-reaching claims, added a hyperparameter-sensitivity analysis, re-computed the perturbation specificity under two additional regulatory priors (DoRothEA A+B+C and DoRothEA all), added a permutation null for the cross-layer PMI “information highway” claim with a consecutive-layer sweep, added a non-tautological characterization of unannotated features using cell-type and CRISPRi tests, added a matched-input cross-model snapshot, added a batch/assay-proxy confound analysis for scGPT, labeled the preliminary scGPT causal-patching result explicitly as preliminary, added justifications for the SVD cosine threshold and Leiden resolution choices, and included graph dimensions, peak memory, and software/hardware versions in Methods. All new analyses are reported in the main text with supporting detail in Additional file 1. The main scientific conclusion is unchanged but now framed more cautiously.

Reviewer 1

R1.1 — TRRUST coverage and cell-type scope of CRISPRi

The perturbation specificity test relies solely on TRRUST covering only 48 TFs, which represents a small and literature-mining-biased subset of known human TF-target relationships. [...] The authors should discuss whether using more comprehensive regulatory network resources (e.g., DoRothEA, ChIP-Atlas, or CellOracle priors) would alter this conclusion. Furthermore, the CRISPRi validation is performed exclusively in K562.

Response: We have re-run the specificity test using DoRothEA (A+B+C confidence and all confidence levels) and adopted a hypergeometric-based “specific” definition ($\text{BH-FDR} < 0.05$, universe = 20,000 human protein-coding genes) so that the numerical comparison is not inflated by DoRothEA’s larger per-TF target sets. Under this stricter definition, specificity rates are 0/28 (0.00%) for TRRUST, 1/12 (8.33%) for DoRothEA A+B+C, and 1/18 (5.56%) for DoRothEA all. The “ ≥ 1 -overlap” definition used in the main-text 6.2% climbs with database size (3.57% TRRUST, 33.33% DoRothEA A+B+C, 38.89% DoRothEA all) but this is an artifact of the statistical test,

removed by the hypergeometric correction. The single-digit rate is therefore robust to database choice. New Results paragraph in §2.9 and new Additional file 1: Table S16. We also explicitly note the K562-only limitation in the Limitations paragraph and in the softened Discussion (§3). A second-cell-line experiment (e.g. RPE1) is listed as the natural next step but is beyond the scope of this revision.

R1.2 — Hyperparameter ablation (expansion $\times k$)

The entire analysis uses a uniform $4\times$ overcomplete dictionary with TopK sparsity $k=32$ across all layers and both models, yet no justification for these specific values is provided. [...] The authors should conduct an ablation study on at least one representative layer.

Response: We added a 9-cell grid of SAEs at Geneformer L11 over expansion $\in \{2\times, 4\times, 8\times\}$ and $k \in \{16, 32, 64\}$, reporting variance explained, dead-feature count, mean decoder cosine similarity, and Leiden module count for each configuration. Results are in new §2.12 of the main text and Additional file 1: Table S14. The qualitative conclusions (substantial superposition, organized biological structure, stable module count) are preserved across the grid; full TF-specificity recomputation per grid cell was beyond the revision’s compute budget.

R1.3 — scGPT causal patching proxy values

As acknowledged in Section 2.7, causal patching for scGPT uses proxy expression values (uniform 1.0 for all genes) rather than true expression inputs, yielding a specificity of only $0.98\times$. This fundamentally undermines fair comparison [...] The authors should either describe a technical path to recover true expression-value inputs for scGPT’s causal analysis, or explicitly label the scGPT causal results as preliminary/inconclusive rather than presenting them in parallel with Geneformer’s results.

Response: We have re-labeled the scGPT causal-patching number in §2.7 as *preliminary / exploratory* and explicitly flagged that the input distribution was substantially out-of-distribution relative to training. We have also dropped it from the side-by-side cross-model presentation. The path to recover true per-cell binned value inputs is described in the new Methods “Input handling for each model” paragraph, and the updated results are reported in new §2.11 of the main text and Additional file 1: Table S17.

R1.4 — Cross-layer PMI permutation null and consecutive-layer sweep

Under a TopK constraint with $k=32$, co-activation across layers has a non-trivial statistical baseline, and the current analysis does not adequately rule out that these cross-layer connections are simply artifacts of the sparsity constraint. A permutation-based null model for PMI thresholding is needed. Additionally, the analysis examines only three layer-pair jumps [...] a consecutive layer-by-layer analysis would better demonstrate whether information propagation is truly continuous or concentrated at specific transitions.

Response: We added a permutation null on the three long-range pairs (20 column-shuffle rounds each) and recomputed the highway fraction at the data-driven threshold $\tau = \max(3.0, \tau_{\text{null}}^{p99})$. We

also added a consecutive-layer sweep for all $L_i \rightarrow L_{i+1}$ pairs at a fixed $\tau=3.0$. The null threshold τ_{null}^{p99} was below 3.0 for every long-range pair tested, so the fixed threshold is already conservative and the null correction leaves the highway fractions unchanged (this is a *negative* null result). We cite Wang et al. (scDrugMap, 2025) in the cross-layer section. New Figure panel and Additional file 1: Tables S18 and S19.

R1.5 — Unannotated feature tests beyond module co-membership

Section 2.11 attempts to address unannotated features through guilt-by-association analysis, but merely showing that they reside within co-activation modules (which cover 96% of all features) is nearly tautological. The authors should further characterize these features by asking: how many exhibit consistent cell-type specificity? How many show significant responses in the CRISPRi perturbation data?

Response: We agree the module-membership test is nearly tautological and now say so in §2.14. We replaced it with two non-tautological tests at Geneformer L0, L5, L11, L17: (i) cell-type specificity in Tabula Sapiens; (ii) perturbation responsiveness against the 100 Replogle CRISPRi targets. The cell-type test did not separate the two populations (both are at the $\sim 90\text{--}95\%$ any-feature ceiling). The perturbation test does separate them: at layer 11, 2.32% of unannotated features (~ 47 features) appear in at least one perturbation target’s top-20 responding-features list with $|\text{effect}| > 0.5$, versus only 0.15% of annotated features (~ 4 features)—a $\sim 15\times$ gap, providing non-tautological evidence that a subset of unannotated features carry real perturbation-response signal. We name Wang et al. (HECLIP, 2025) as a recommended orthogonal future validation in the main text. New Additional file 1: Table S20.

Reviewer 2

R2 Major 1 — scGPT proxy values

The author should provide a rigorous justification for using 1.0 as a substitute, or rerun the experiments to compare these results with those obtained using the actual continuous expression levels.

Response: See response to R1.3. The scGPT causal-patching number with the uniform-1.0 proxy is now explicitly labeled preliminary and excluded from the cross-model comparison.

R2 Major 2 — SVD cosine threshold

The author should clarify the rationale behind selecting > 0.7 as the threshold. Please provide relevant references to support this choice, or include a brief ablation study.

Response: We discovered that the paper text quoted $\tau=0.7$ while the pipeline actually used $\tau=0.5$; we have corrected the main text accordingly. We added a sensitivity analysis over $\tau \in \{0.3, 0.5, 0.7, 0.9\}$ at all 18 Geneformer layers (new Additional file 1: Table S11). At the correct

$\tau=0.5$, 0.3% of features on average are SVD-aligned, reproducing the per-layer counts in Table 1. Across the whole range of τ , >98% of ontology enrichments live exclusively in the non-aligned set, so the headline conclusion that SAE features carry the biological signal is robust.

R2 Major 3 — PMI graph dimensions, memory, and Leiden resolution

I suggest the author explicitly state the dimensions of the constructed graphs and detail the memory/VRAM consumption required for this step. Furthermore, please explain the biological or computational rationale for using the default parameter of 1.0 for the Leiden community detection resolution.

Response: We now report N_{feat} , number of edges after the significance threshold, coactivation-matrix memory, and peak resident set size per layer in Additional file 1: Table S12 (new). For Leiden resolution, we followed the `leidenalg` / `scanpy` default $\gamma=1.0$; to quantify sensitivity, we swept $\gamma \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$ at Geneformer L0 and L11 and report module count, mean module size, coverage, and ARI vs. the $\gamma=1.0$ baseline (new Additional file 1: Table S13). The partition is nested rather than uniquely stable (ARI between $\gamma=1.0$ and $\gamma=0.75/1.5$ is 0.17–0.26), but the qualitative module identities (cell cycle, translation, metabolism, ...) are preserved across the range. We have rewritten the main-text wording to reflect this accurately.

R2 Major 4 — scGPT perturbation mapping

I suggest the author supplement this study by performing the same perturbation response mapping experiments on scGPT to verify whether it exhibits similarly low specificity.

Response: We ran a lean scGPT perturbation response mapping on Replogle K562 at layer 7 (50 targets, 10 cells per target, 100 non-targeting controls, per-cell binned expression values re-tokenized from the Replogle source h5ad). scGPT exhibits the same qualitative regime as Geneformer: single-digit loose overlap ($2/28 = 7.14\%$ TRRUST, $2/12 = 16.67\%$ DoRothEA A+B+C), and $0/28 / 0/12$ under hypergeometric FDR correction. Reported in §2.9 and Additional file 1: Table S15. This cross-validates the central Geneformer finding across both architecturally distinct scFMs.

R2 Minor 1 — Language polish

Response: We have done a language pass, tightening Background and Conclusions.

R2 Minor 2 — Force-directed figure captions

Response: Additional file 1: Figures S2 and S3 captions now explicitly state “Fruchterman–Reingold force-directed layout of the intra-module co-activation graph; this is *not* a UMAP or t-SNE projection”.

R2 Minor 3 — Software and hardware details

Response: We added a new Methods subsection “Computational environment” listing software stack (PyTorch + MPS, NumPy, scanpy, leidenalg, decoupler-py, ...), the Apple Silicon hardware, macOS version, and wall-clock time for the longest steps (extraction, SAE training, PMI + Leiden, perturbation mapping). Exact package versions are pinned in the code repository `requirements.txt`.

R2 Minor 4 — Single-command reproducibility

Response: We added `run_all.sh` and `notebooks/walkthrough.ipynb` to the code repository. The shell script chains the Phase 1 → Phase 2 → Phase 3 steps; the notebook runs one layer end-to-end on a small subsample for reviewers who do not want to commit 400 GB of disk.

Reviewer 3

R3.1 — Factual inaccuracies about scGPT architecture

scGPT uses binned gene expression values, not continuous expression values during pretraining (line 111) [...] In Table 3, “continuous expression” should be “binned expression,” and the training objectives are swapped.

Response: Corrected in all locations. Line 162 now reads “dual input of gene-identity tokens plus binned expression-value embeddings (51 bins by default)”; Table 3 now reads “Gene token + binned value” / “Generative (causal attention)” for scGPT and “Rank-value tokens” / “Masked-gene modelling” for Geneformer. A new Methods paragraph “Input handling for each model” gives the full description of how each model ingests a cell.

R3.2 — Cross-model comparison confound

The cross-model comparison does not seem well controlled. Geneformer activations are extracted from baseline K562 cells, while scGPT activations are extracted from Tabula Sapiens cells.

Response: We have added an explicit caveat to §2.3 and to the caption of Figure 3, now stating that the observed cross-model differences conflate architecture with input distribution and should be read qualitatively. We also added a partial matched-input control (§2.13, Additional file 1: Table S21) in which K562-trained Geneformer SAEs are applied to 3,000 Tabula Sapiens cells. Variance explained drops by 18–23 percentage points at L0, L5, and L11 when Geneformer sees Tabula Sapiens data instead of K562 data, so the 9-percentage-point Geneformer-vs.-scGPT gap in Table 3 is a lower bound on the true architectural gap; under matched input, Geneformer trails scGPT by ~27 percentage points. A fully matched retraining is noted as the natural follow-up.

R3.3 — SAEs trained on control/baseline data

I am not convinced that SAEs trained on normal or control-state data are the right setup for probing transcription factor regulatory logic. [...] If the training data are not perturbed [...] it is unclear why one should expect the learned features to recover TF-target regulatory logic in a strong sense.

Response: We agree this is a fundamental caveat and have added a dedicated Discussion paragraph §3 that states this explicitly: SAEs trained on unperturbed K562 controls and unperturbed Tabula Sapiens tissue are structurally biased toward cell-state / co-expression features, and it is *a priori* unclear that such SAEs should recover TF→target regulatory wiring in a strong sense. We now describe, as the clearest single next experiment, training SAEs directly on pooled perturbed + control activations with perturbation identity as an auxiliary signal.

R3.4 — “Minimal regulatory logic” overclaim

I also do not think the K562+TabSap control is sufficient to isolate the model itself as the bottleneck. [...] the claims that the models encode only “minimal regulatory logic” are too strong as currently written.

Response: We have softened the claims across the Abstract, §2.10, Discussion (§3), and Conclusions. The multi-tissue result is now framed as “consistent with, but not proving, a model-level limitation,” and three alternative explanations are listed: (i) the pooling strategy is too simple, (ii) baseline-only SAE training biases away from TF wiring, (iii) a different SAE objective (perturbation-supervised) could behave differently. The title and headline number remain but their scope is now explicitly “under the conditions tested.”

R3.5 — Batch / assay features

Do the SAEs recover batch specific or assay specific features? This seems especially relevant for Tabula Sapiens, which includes multiple sequencing assays and cross donor variations.

Response: We added a dedicated “Batch and assay confounds” subsection (§2.18) and a new Additional file 1: Table S22. For the scGPT atlas at layers {0, 4, 7, 11} we compute, per feature, the mutual information between its activation and (a) the tissue label (our proxy for sample-level batch, because the cached scGPT extraction metadata carries only {tissue, cell_type}, not donor or assay IDs) and (b) the cell-type label. Mean MI with cell type is $\sim 2.5\times$ larger than mean MI with tissue across all four layers, and essentially zero features are tissue-dominant (0/1990 at L0, 0/2038 at L4, 0/1999 at L7, 1/1766 at L11). A full donor-level / assay-level analysis requires joining the Tabula Sapiens source h5ad on `cell_idx` and is listed as a follow-up.

R3.6 — Finetuned scGPT on Replogle

Would model finetuning help? In particular, it would be interesting to test a perturbation task tuned scGPT checkpoint on Replogle dataset to see if it recovers more signal.

Response: We agree that a perturbation-finetuned scGPT checkpoint is the single most promising

next experiment, and have named it explicitly in the Discussion as a follow-up. We have not run it in this revision because it requires both the fine-tuned checkpoint and a re-run of the full SAE pipeline; it is flagged as the clearest test of the bottleneck question.

Summary of changes

1. Factual corrections: scGPT input encoding (continuous $\rightarrow$ binned, 51 bins); Geneformer/scGPT training objectives swapped (Geneformer is masked-token, scGPT is generative); SVD threshold text (0.7 $\rightarrow$ 0.5, matching the pipeline).
2. New Results subsections: §2.11 scGPT causal patching with true binned values; §2.12 hyperparameter ablation at L11; §2.13 matched-input cross-model snapshot; §2.18 batch and assay confounds.
3. New Additional file 1 tables: Tables S11 (SVD sensitivity), S12 (PMI graph / memory), S13 (Leiden sweep), S14 (hyperparameter ablation), S15 (scGPT perturbation test), S16 (multi-database perturbation), S17 (scGPT causal, true values), S18 and S19 (null-corrected and consecutive highways), S20 (unannotated feature tests), S21 (matched-input comparison), and S22 (batch / assay). The original Tables S1–S10 are retained.
4. Softened claims in Abstract, §2.10, Discussion, Conclusions; new explicit “bottleneck” caveat paragraph (§3).
5. New Methods subsections: input handling per model; SVD threshold justification + sweep; PMI graph dimensions + memory; Leiden resolution justification + sweep; computational environment.
6. Figure S2–S3 captions updated to explicitly state force-directed layout.
7. Code repository additions: `run_all.sh`, `notebooks/walkthrough.ipynb`, and all new revision-phase analysis scripts under `src_revision/`.

We trust this revision addresses all three reviewers’ concerns. We are happy to provide any additional analysis needed.

Sincerely,
Ihor Kendiukhov